Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
A		2628850

Battery Evolution: The Physics of Coupled States

Smartphone battery behavior is difficult to model accurately because it emerges from the interaction between user behavior, hardware power demand, and nonlinear electrochemical constraints. Many existing approaches either rely on black-box prediction without physical interpretability, simplify battery dynamics to linear energy balance, or treat power consumption as a single aggregated quantity, thereby obscuring the mechanisms governing discharge under different usage conditions.

In this work, we develop a **physics-based, continuous-time modeling framework** for smartphone battery discharge that preserves causal structure and electrical realism. The battery is represented by a **closed-loop, coupled system of ordinary differential equations (ODEs)** describing the joint evolution of **state of charge (SOC)**, **terminal voltage dynamics**, **thermal effects**, and **capacity degradation**. An explicit **power–current–voltage coupling** is incorporated, ensuring that current demand is endogenously determined by both load power and battery state. This formulation naturally captures the **nonlinear discharge acceleration and voltage-limited shutdown behavior** observed at low SOC.

To connect user behavior with battery dynamics, we construct a **component-level power decomposition model** that disaggregates total power consumption into contributions from the **display, processor, network communication, navigation (GPS), background processes, and system overhead**. Each component is modeled using physically and empirically motivated scaling relationships, allowing diverse usage patterns to be represented consistently within a unified framework.

The proposed model is evaluated through **numerical simulation**, **stochastic Monte Carlo extension**, and **one-at-a-time sensitivity analysis**. Monte Carlo simulations characterize realistic variability in user behavior and operating conditions, producing distributions of battery endurance rather than single-point estimates. Sensitivity analysis identifies the parameters and subsystems that most strongly influence time-to-empty, clarifying the relative roles of computational workload, connectivity, internal resistance, and aging effects.

Overall, this study provides a **white-box, interpretable, and generalizable framework** for understanding smartphone battery behavior. By combining physical battery modeling with component-level power attribution and uncertainty analysis, the model offers explanatory insight and supports informed user guidance and operating-system-level power management strategies.

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1 Introduction

The pervasive integration of smartphones into daily life has rendered them indispensable tools for communication, productivity, and entertainment. This reliance is underpinned by sustained battery performance, which has consequently emerged as a critical determinant of overall user satisfaction and device utility. As smartphone functionalities continue to expand and usage intensity grows, the demands placed on battery systems have escalated significantly. Challenges such as limited operational duration per charge cycle and the necessity for frequent recharging not only impair user convenience but also pose long-term risks to battery health and longevity.

Battery depletion rates are highly contingent upon specific usage patterns. **Resource-intensive applications**—including high-fidelity gaming, multimedia streaming, and navigation—impose substantial energy loads, leading to rapid discharge. In contrast, periods of inactivity or low-engagement tasks such as voice calls and messaging exhibit markedly lower power consumption. Beyond usage profiles, extrinsic and intrinsic factors including operational ambient temperature and progressive battery degradation due to chemical aging further modulate performance metrics and efficiency.

To address these complexities, a **systematic and predictive analytical framework** is required. This necessitates the development of robust mathematical models capable of simulating battery dynamics under heterogeneous conditions. By employing **continuous-time differential equations**, it becomes possible to quantitatively assess the influence of various parameters on discharge characteristics and overall battery behavior. Such modeling not only facilitates accurate performance forecasting across diverse usage scenarios but also establishes a foundation for formulating optimized, intelligent battery management protocols. Ultimately, this research seeks to derive **actionable insights** that can prolong functional battery lifespan, mitigate performance degradation, and elevate the quality of the user experience through informed, adaptive power management strategies.

1.1 The break down of questions

1. **Modeling and Prediction**
   Develop a continuous-time mathematical model to describe battery behavior. This model will be used to predict battery performance in different scenarios, including the calculation of key state parameters (e.g., real-time charge level) and the evaluation of overall performance metrics.

2. **Optimization and Attribution Analysis**
   Focus on optimizing battery performance to improve energy efficiency. Core tasks involve predicting the time-to-empty (battery depletion) and conducting an in-depth analysis of how various factors—such as usage patterns and environmental conditions—specifically impact battery lifespan.

3. **Impact Analysis and Strategy Design**
   Investigate the influence of battery usage behavior on specific external metrics (e.g., sleep quality). Based on this analysis, determine optimal usage conditions and design corresponding battery regulation strategies.

4. **Comprehensive Evaluation and Solution Provision**
   Conduct a comprehensive evaluation of the effectiveness of the proposed regulation strategies.

Analyze the sensitivity of battery performance to different parameters. Finally, synthesize the research findings to provide users with an optimal battery management solution.

2 Analytical Approach and Logical Framework

2.1 Logical Framework

Our analysis follows a four-phase, closed-loop pipeline that integrates deterministic modeling, stochastic simulation, sensitivity analysis, and actionable insight generation. This ensures that results are physically grounded, robust to variability, and directly translatable into user guidance.

1. **Deterministic Physics-Based Baseline**
   We construct a continuous-time model from equivalent circuit theory, capturing the battery's core states: State of Charge (SOC, $\xi(t)$) and terminal voltage ($V_{\text{term}}(t)$):

   $$V_{\text{term}} = U_{\text{ocv}}(\xi) - I \cdot R_{\text{int}}(\xi, T), \quad \frac{d\xi}{dt} = -\frac{P_{\text{batt}}}{E_{\text{eff}}}.$$

   Here, E_{eff} denotes the **effective usable energy** of the battery, which is approximately equal to the product of the **effective usable capacity** and the **average operating voltage**:

   $$E_{\text{eff}} \approx C_{\text{eff}} \cdot \bar{V}. \tag{1}$$

   Device power demand $P_{\text{total}}(t)$ is decomposed into idle, screen, CPU, network, GPS, and background components. This yields a deterministic prediction of Time-to-Empty (TTE_0) and SOC/voltage trajectories, providing an initial energy attribution snapshot.

2. **Monte Carlo Simulation of Real-World Variability**
   Key behavioral parameters (e.g., high-CPU task duration, network usage) are treated as random variables. Repeated sampling generates distributions of usage trajectories, producing a probabilistic range of TTE (e.g., 3.5–4.2 h), which captures realistic uncertainty.

3. **Parameter Sensitivity and Dominant Driver Analysis**
   Each critical parameter (e.g., base CPU load, screen brightness, internal resistance) is perturbed (±10%) and the effect on mean TTE (ΔTTE) is computed. Dominant drivers are identified and visualized with a tornado diagram, highlighting which factors most strongly influence battery life.

4. **Synthesis into Practical Strategies**
   Insights inform targeted user recommendations (e.g., reducing brightness is twice as effective as disabling Bluetooth) and system-level power-saving strategies. Battery aging effects (increase in R_{int}, decrease in E_{eff}) are incorporated, and the framework is generalizable to other portable devices.

This four-phase cycle closes the loop: physics-based modeling provides causality, stochastic simulation tests variability, sensitivity analysis identifies key drivers, and synthesis converts understanding into actionable guidance.

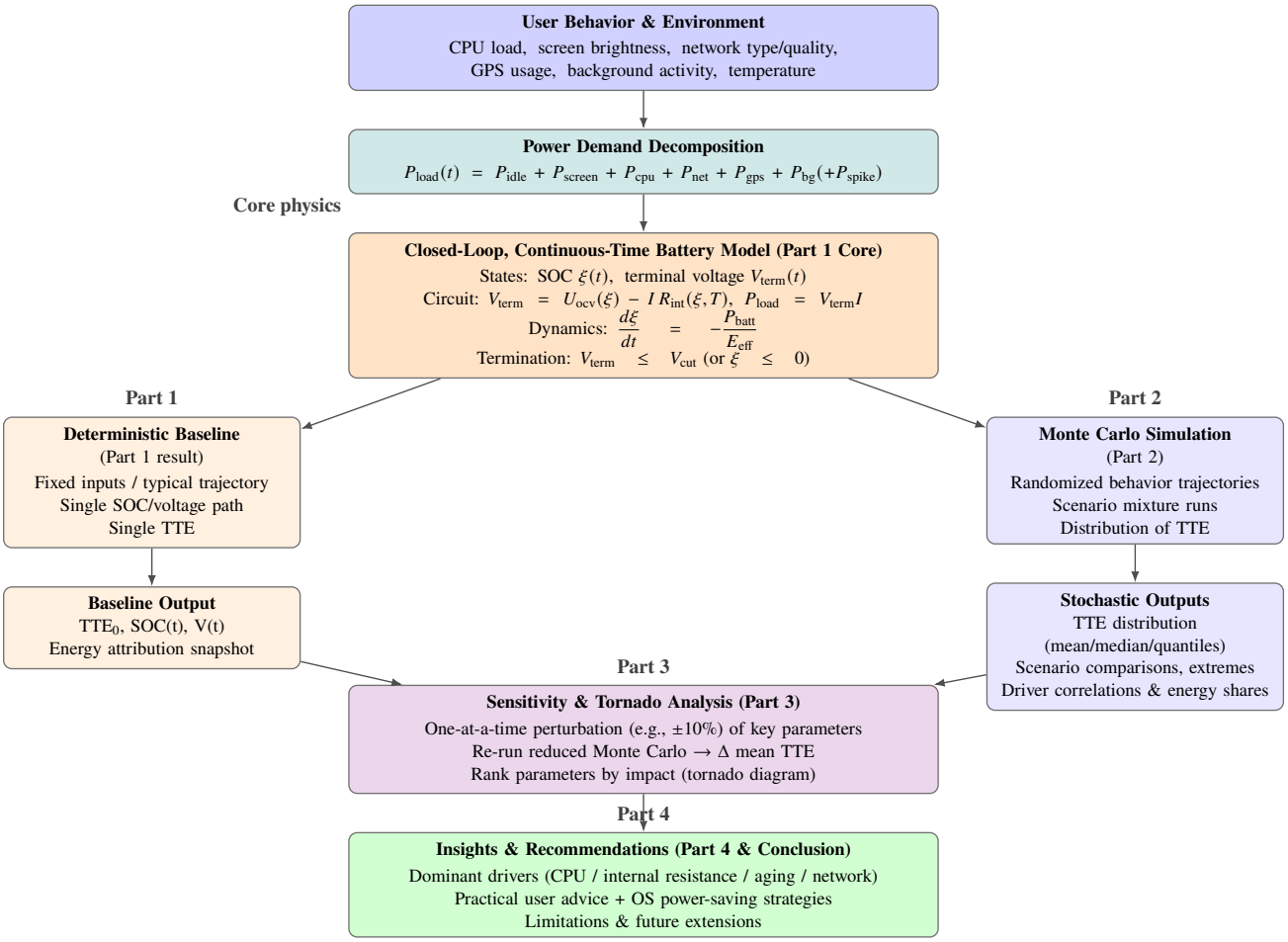

Figure 1: Paper workflow and model pipeline overview.

2.2 Model Assumptions

To simplify the problem while preserving the operability and predictive accuracy of the proposed model, several assumptions are adopted in this study.

Assumption 1 (Temperature Dependence). The electrochemical characteristics of the battery at different operating temperatures can be effectively represented by a mathematical model. The influence of temperature on battery performance is primarily manifested through variations in internal impedance and electrochemical reaction rates.

Assumption 2 (Capacity Degradation). Battery capacity degradation over time is assumed to follow an exponential decay pattern. The aging process is gradual and progressive, and abrupt or stepwise performance failures are excluded from consideration.

Assumption 3 (Power Consumption Statistics). Under different usage scenarios, the battery power consumption exhibits stable statistical properties. These usage-dependent power profiles can be accurately estimated through experimental measurements and subsequent data-fitting procedures.

Assumption 4 (Voltage Simplification). During normal operation, the terminal voltage of the battery remains within a relatively narrow range. Consequently, a constant average voltage can be adopted to simplify the model formulation without introducing significant loss of accuracy. Note

that assumption 4 applies only to short-term power estimation, while voltage dynamics are explicitly resolved in the closed-loop battery model.

3 Nomenclature of battery parameters

Table 1: Nomenclature of main symbols.

Symbol	Description	Unit
$A_i(t)$	Activity level of background task i	–
$B(t)$	Screen brightness level (normalized)	–
C_{nom}	Nominal battery capacity	Ah
C_p	Polarization capacitance	F
C_{th}	Thermal capacity	J/K
$C_{\text{eff}}(t,T)$	Effective battery capacity	Ah
C_{usable}	Usable battery capacity	Ah
E_a	Activation energy for aging	J/mol
$F_{\text{nav}}(t)$	GPS sampling frequency	Hz
$I(t)$	Discharge current	A
$L(t)$	Link loss factor	–
N_0	Noise spectral density	dBm/Hz
P	Power	W
P_{bb}	Baseband power	W
$P_{\text{bg}}(t)$	Background power	W
$P_{\text{brightness}}(t)$	Screen display power	W
$P_{\text{cpu}}(t)$	Processor power	W
$P_{\text{idle}}(T)$	Base system power	W
$P_{\text{nav}}(t)$	Navigation (GPS) power	W
$P_{\text{net}}(t)$	Network communication power	W
$P_{\text{other}}(t)$	Other auxiliary power	W
$P_{\text{total}}(t,s)$	Total electrical power demand	W
$P_{\text{tx}}(t)$	Transmit power	W
$Q_{\text{gen}}(t)$	Internal heat generation	W
R	Universal gas constant	J/(mol·K)
$R_0(\xi,T)$ or $R(\xi,T)$	Internal resistance	Ω
R_p	Polarization resistance	Ω
$S(t)$	Screen on/off state (0 or 1)	–
$T(t)$	Battery temperature	K
T_{amb}	Ambient temperature	K
$U(t)$	CPU utilization (normalized)	–
V_{cut}	Cutoff voltage	V
$V_{\text{dd}}(t)$	CPU supply voltage	V
$V_{\text{ocv}}(\xi)$	Open-circuit voltage	V
$V_p(t)$	Polarization voltage	V
$V_{\text{term}}(t)$	Terminal voltage	V

Continued on next page

Symbol	Description	Unit
W	Communication bandwidth	Hz
c_0, c_1, c_2, c_3	Coefficients for CPU power polynomial	W
$f(t)$	CPU frequency	Hz
$f_{\text{age}}(t, T)$	Aging factor	–
h	Convective heat transfer coefficient	W/K
k_0	Pre-exponential factor for aging	s^{-1}
$q(t)$	Remaining charge	Ah
t	Time	s
δ	Exponent for brightness scaling	–
η	Coulombic efficiency	–
η_{PA}	Power amplifier efficiency	–
$\xi(t)$	State of charge (SOC)	–

4 Physics-Based Continuous-Time Battery Model

4.1 State-of-charge definition and charge balance

We define the state of charge (SOC) $\xi(t)$ as the fraction of remaining charge $q(t)$ relative to the nominal rated capacity C_{nom}:

$$\xi(t) \equiv \frac{q(t)}{C_{\text{nom}}}, \quad \xi(t) \in [0, 1], \tag{2}$$

where $q(t)$ is measured in ampere-hours (Ah). When $\xi = 1$, the battery is fully charged; when $\xi = 0$, it is fully discharged in the ideal (lossless) sense.

During discharge, the instantaneous electrical power demanded by the smartphone satisfies

$$P(t) = V_{\text{term}}(t)\, I(t), \tag{3}$$

where $V_{\text{term}}(t)$ is the terminal voltage and $I(t) > 0$ denotes the discharge current.

Charge conservation gives

$$\frac{dq(t)}{dt} = -I(t), \tag{4}$$

hence the ideal SOC dynamics are

$$\frac{d\xi(t)}{dt} = -\frac{I(t)}{C_{\text{nom}}}. \tag{5}$$

4.1.1 Coulombic efficiency and effective usable capacity

In practical lithium-ion batteries, not all nominal capacity is electrochemically accessible due to irreversible processes such as SEI growth and side reactions. We represent this effect using a coulombic usability factor $\eta \in (0, 1]$, interpreted as the fraction of nominal capacity available for reversible discharge. The corresponding usable capacity is

$$C_{\text{usable}} = \eta\, C_{\text{nom}}. \tag{6}$$

4.1.2 Capacity degradation and aging modulation

Long-term battery aging further reduces usable capacity through cycle aging and calendar aging, both of which are temperature dependent. We introduce an aging factor $f_{\text{age}}(t,T) \in (0,1]$ such that the effective usable capacity is

$$C_{\text{eff}}(t,T) = C_{\text{nom}} \cdot \eta \cdot f_{\text{age}}(t,T). \tag{7}$$

Over a single discharge episode (hours), $f_{\text{age}}(t,T)$ evolves slowly compared with $\xi(t)$. Accordingly, its primary role is to scale the effective capacity available during discharge. The SOC dynamics are therefore modeled as

$$\frac{d\xi(t)}{dt} = -\frac{I(t)}{C_{\text{nom}} \cdot \eta \cdot f_{\text{age}}(t,T(t))}. \tag{8}$$

4.1.3 Terminal voltage and equivalent circuit model

A Thevenin-type equivalent circuit is adopted:

$$V_{\text{term}}(t) = V_{\text{ocv}}(\xi(t)) - I(t)\, R_0(\xi(t),T(t)) - V_p(t), \tag{9}$$

where $V_{\text{ocv}}(\xi)$ is the SOC-dependent open-circuit voltage, $R_0(\xi,T)$ denotes the effective internal resistance, and $V_p(t)$ represents the polarization voltage accounting for transient electrochemical effects.

To represent the characteristic plateau–cliff structure of the OCV–SOC curve while maintaining smoothness, the open-circuit voltage is modeled using a low-order trigonometric expansion:

$$V_{\text{ocv}}(\xi) = \sum_{i=1}^{3} a_i \sin(b_i \xi + c_i)\,, \tag{10}$$

with coefficients calibrated to preserve a physically plausible and monotonic OCV shape over $\xi \in [0,1]$.

4.1.4 Power–current–voltage coupling

The total electrical power demanded by the smartphone is

$$P_{\text{total}}(t,s) = I(t)\, V_{\text{term}}(t), \tag{11}$$

where s denotes a usage scenario.

Substituting the terminal voltage expression yields

$$P_{\text{total}}(t,s) = I(t)\left[V_{\text{ocv}}(\xi(t)) - V_p(t) - I(t)\, R(\xi(t),T(t))\right]. \tag{12}$$

Rearranging gives a quadratic equation in $I(t)$:

$$R(\xi,T)\, I^2(t) - \big(V_{\text{ocv}}(\xi) - V_p(t)\big)\, I(t) + P_{\text{total}}(t,s) = 0. \tag{13}$$

The physically admissible solution is

$$I(t) = \frac{\big(V_{\text{ocv}}(\xi) - V_p(t)\big) - \sqrt{\big(V_{\text{ocv}}(\xi) - V_p(t)\big)^2 - 4R(\xi,T)\, P_{\text{total}}(t,s)}}{2R(\xi,T)}. \tag{14}$$

For feasibility, the discriminant must satisfy

$$\big(V_{\text{ocv}}(\xi) - V_p(t)\big)^2 \geq 4R(\xi,T)\, P_{\text{total}}(t,s), \tag{15}$$

otherwise the demanded power exceeds what the battery can deliver at the current state, leading to voltage collapse, power limiting, or device shutdown.When the feasibility condition is violated, i.e.,

$$\left(V_{\text{ocv}}(\xi) - V_p(t)\right)^2 < 4R(\xi, T)\, P_{\text{total}}(t, s), \tag{16}$$

the battery is unable to sustain the requested power at the current state. In practice, smartphones employ power limiting mechanisms to prevent voltage collapse.

Accordingly, we impose an instantaneous power cap

$$P_{\max}(t) = \frac{\left(V_{\text{ocv}}(\xi) - V_p(t)\right)^2}{4R(\xi, T)}, \tag{17}$$

and replace the demanded power by

$$P_{\text{eff}}(t, s) = \min\{P_{\text{total}}(t, s),\ P_{\max}(t)\}\,. \tag{18}$$

The effective power P_{eff} is then used in the coupled power–current relation to ensure numerical feasibility and physical consistency.

4.1.5 Closed-loop SOC evolution and end-of-discharge condition

Combining the SOC dynamics with the coupled current expression yields the closed-loop ODE

$$\frac{d\xi(t)}{dt} = -\frac{1}{C_{\text{nom}}\,\eta\, f_{\text{age}}(t, T(t))} \cdot \frac{\left(V_{\text{ocv}}(\xi) - V_p(t)\right) - \sqrt{\left(V_{\text{ocv}}(\xi) - V_p(t)\right)^2 - 4R(\xi, T)\, P_{\text{total}}(t, s)}}{2R(\xi, T)}. \tag{19}$$

Battery depletion is defined to occur when the terminal voltage drops below a cutoff value V_{cut}:

$$V_{\text{term}}(t^*) \le V_{\text{cut}}. \tag{20}$$

This voltage-based termination criterion naturally captures nonlinear end-of-discharge behavior without imposing ad hoc SOC thresholds.

4.1.6 Polarization Voltage Dynamics

The polarization voltage $V_p(t)$ is modeled using a first-order RC branch, representing the transient voltage drop induced by charge-transfer kinetics and diffusion processes within the battery. Its temporal evolution is governed by the following ordinary differential equation:

$$\frac{dV_p(t)}{dt} = -\frac{1}{R_p C_p} V_p(t) + \frac{1}{C_p} I(t), \tag{21}$$

where R_p denotes the polarization resistance, which is generally a function of the state of charge and temperature, C_p is the polarization capacitance, and $I(t)$ is the discharge current.

This formulation explicitly captures the voltage relaxation behavior and hysteresis effects commonly observed in lithium-ion batteries, thereby enhancing the physical interpretability and dynamic fidelity of the proposed model.

4.2 Thermal Dynamics

Battery performance is strongly influenced by thermal effects, as both electrochemical kinetics and capacity degradation are temperature dependent. To capture these phenomena in a continuous-time framework, we incorporate coupled thermal dynamics and Arrhenius-type aging.

4.2.1 Temperature-dependent capacity fade

The effective usable capacity is defined as

$$C_{\text{eff}}(t,T) = C_{\text{nom}} \cdot \eta \cdot f_{\text{age}}(t,T), \tag{22}$$

where $f_{\text{age}}(t,T) \in (0,1]$ denotes the remaining fraction of usable battery capacity.

The evolution of capacity fade is modeled using an Arrhenius-type law:

$$\frac{df_{\text{age}}(t,T)}{dt} = -k_0 \exp\left(-\frac{E_a}{R\,T(t)}\right), \tag{23}$$

where k_0 is a pre-exponential degradation rate, E_a is the activation energy, and R is the universal gas constant.

In this work, additional dependencies on state of charge and polarization voltage are neglected, as capacity aging occurs on a much longer time scale than discharge dynamics. Accordingly, the modulation function $g(\cdot)$ is set to unity.

4.2.2 Thermal energy balance

The battery temperature evolves according to

$$\frac{dT(t)}{dt} = \frac{1}{C_{\text{th}}}\left[Q_{\text{gen}}(t) - h\big(T(t) - T_{\text{amb}}\big)\right], \tag{24}$$

where C_{th} is the thermal capacity, h the convective heat transfer coefficient, and T_{amb} the ambient temperature. Internal heat generation is dominated by Joule heating:

$$Q_{\text{gen}}(t) \approx I^2(t)\, R\big(\xi(t), T(t)\big). \tag{25}$$

4.2.3 Coupled SOC–thermal–aging dynamics

The resulting coupled system is

$$\begin{cases} \dfrac{d\xi(t)}{dt} = -\dfrac{I(t)}{C_{\text{nom}} \cdot \eta \cdot f_{\text{age}}(t,T(t))}, \\ \dfrac{dV_p(t)}{dt} = -\dfrac{1}{R_p C_p}\, V_p(t) + \dfrac{1}{C_p}\, I(t), \\ \dfrac{df_{\text{age}}(t,T)}{dt} = -k_0 \exp\left(-\dfrac{E_a}{RT(t)}\right), \\ \dfrac{dT(t)}{dt} = \dfrac{1}{C_{\text{th}}}\Big[I^2(t)R(\xi(t),T(t)) - h(T(t) - T_{\text{amb}})\Big]. \end{cases} \tag{26}$$

The parameters in (21) take values within physically reasonable ranges for lithium-ion smartphone batteries. Specifically, the polarization resistance R_p typically lies in the range

$$R_p \in [0.05,\ 0.2]\ \Omega,$$

while the polarization capacitance C_p usually falls within

$$C_p \in [1000,\ 5000]\ \text{F}.$$

These ranges are consistent with commonly reported values in equivalent circuit models for consumer-grade lithium-ion cells and reflect the short-term transient response characteristics of the battery during discharge.

Equations (21–22) constitute a physically consistent, continuous-time, coupled battery model incorporating electrochemical, thermal, and degradation dynamics. For the scope of this study, we assume a fixed convective environment (T_{amb} and h constant), allowing for a tractable yet realistic description of thermal effects under typical usage conditions.

4.3 Physical Decomposition of the Power Consumption Model

The total instantaneous power consumption of a smartphone is decomposed into contributions from multiple hardware subsystems. Enforcing energy conservation, the total electrical load is expressed as[3]

$$P(t) = P_{\text{idle}}(T) + P_{\text{brightness}}(t) + P_{\text{cpu}}(t) + P_{\text{net}}(t) + P_{\text{nav}}(t) + P_{\text{bg}}(t) + P_{\text{other}}(t), \tag{27}$$

representing base system power, display power, processor power, network communication, navigation/GPS, background activity, and other auxiliary loads.

4.3.1 Base Power – $P_{\text{idle}}(T)$

The base (idle) power captures operating-system overhead, memory refresh, sensor standby, and static leakage. Its temperature dependence is modeled as

$$P_{\text{idle}}(T) = P_{\text{idle,ref}} \left[1 + k_{\text{temp}} \left(T - T_{\text{ref}}\right)\right]. \tag{28}$$

Parameter estimates for Base Power:

1. $P_{\text{idle,ref}} \approx 0.5 - 1.5$ W
2. $k_{\text{temp}} \approx 0.005 - 0.02$ K^{-1}
3. $T_{\text{ref}} = 298$ K

4.3.2 Screen Display Power – $P_{\text{brightness}}(t)$

The screen power is modeled as a convex function of normalized brightness, reflecting the nonlinear efficiency characteristics of OLED/LCD backlights:

$$P_{\text{brightness}}(t) = \left[P_{\text{screen,min}} + \left(P_{\text{screen,max}} - P_{\text{screen,min}}\right)\left(\frac{B(t)}{B_{\text{max}}}\right)^{\delta}\right] \cdot S(t), \tag{29}$$

where $B(t)$ is the instantaneous brightness level, $S(t) \in \{0, 1\}$ indicates screen on/off state, and $\delta > 1$ induces superlinear scaling.

Parameter estimates for Screen Power:

1. $P_{\text{screen,min}} \approx 0.2 - 0.5$ W
2. $P_{\text{screen,max}} \approx 1.5 - 2.5$ W
3. $\delta \approx 1.5 - 2.0$

4.3.3 Processor Power – $P_{\text{cpu}}(t)$

At the circuit level, processor power consists of dynamic switching power and temperature-dependent leakage:

$$P_{\text{cpu}}(t) = C_{\text{eff}} V_{\text{dd}}^2(t)\, f(t)\, U(t) + V_{\text{dd}}(t)\, I_{\text{leak}}(T), \tag{30}$$

where $U(t) \in [0, 1]$ denotes normalized CPU utilization.

In modern mobile processors, dynamic voltage and frequency scaling (DVFS) introduces strong coupling between $V_{\text{dd}}(t)$, $f(t)$, and workload. Over the operating envelope of interest, this relationship is well approximated by an empirical superlinear polynomial in utilization:

$$P_{\text{cpu}}(t) \approx c_0 + c_1 U(t) + c_2 U^2(t) + c_3 U^3(t), \tag{31}$$

which captures both leakage-dominated idle behavior and rapidly increasing power draw under high computational load.

Parameter estimates for Processor Power:
1. Static leakage ($U = 0$): $\approx 0.2 - 0.5$ W
2. Dynamic peak ($U \approx 1$): $\approx 3 - 6$ W
3. Polynomial coefficients c_i fitted over $0 \leq U \leq 1$

4.3.4 Network Communication Power – $P_{\mathbf{net}}(t)$

Network power consumption consists of radio-frequency transmission and baseband processing:

$$P_{\text{net}}(t) = \frac{P_{\text{tx}}(t)}{\eta_{\text{PA}}} + P_{\text{bb}}. \tag{32}$$

The required transmit power is modeled using a Shannon-type expression augmented by an effective link-loss factor $L(t) \geq 1$, capturing path loss, shadowing, and implementation margins:

$$P_{\text{tx}}(t) = L(t)\, N_0\, W \left(2^{\frac{R(t)}{W}} - 1\right). \tag{33}$$

Parameter estimates for Network Power:
1. Bandwidth W: Wi-Fi 20–160 MHz, 5G NR 100–400+ MHz
2. Noise spectral density $N_0 \approx -174$ dBm/Hz
3. Power amplifier efficiency $\eta_{\text{PA}} \approx 0.2 - 0.4$
4. Baseband power $P_{\text{bb}} \approx 0.2 - 1$ W
5. Effective link-loss factor $L(t)$ captures signal quality variation
6. Total network power: Wi-Fi 0.2–0.8 W, 4G/5G 0.5–2.0 W

4.3.5 Navigation (GPS) Power – $P_{\mathbf{nav}}(t)$

Navigation power is modeled as

$$P_{\text{nav}}(t) = \begin{cases} P_{\text{nav,base}} \dfrac{F_{\text{nav}}(t)}{F_{\text{nav,ref}}}, & \text{if GPS active,} \\ 0, & \text{otherwise,} \end{cases} \tag{34}$$

where $F_{\text{nav}}(t)$ is the GNSS sampling frequency.

Parameter estimates for Navigation Power:
1. $P_{\text{nav,base}} \approx 0.2 - 0.6$ W
2. $F_{\text{nav,ref}} = 1$ Hz
3. Multi-frequency GNSS increases power by 30–40%

4.3.6 Background Application Power – $P_{\mathbf{bg}}(t)$

Aggregated background power from multiple concurrent tasks is modeled as[4]

$$P_{\text{bg}}(t) = \sum_{i=1}^{N} A_i(t)\, P_{\text{task},i}, \tag{35}$$

where $A_i(t) \in [0, 1]$ denotes task activity and $P_{\text{task},i}$ its nominal power draw.

Parameter estimates for Background Power:
1. Single task: 0.05 – 0.5 W
2. Multi-task aggregation: 0.2 – 1.0 W

4.3.7 Parameter Values

Component	Typical Range (W)	Notes
Base Power	0.5 – 1.5	System idle + static leakage
Screen Display	0.2 – 2.5	Brightness dependent
CPU	0.3 – 6	Load & DVFS dependent
Network (Wi-Fi)	0.2 – 0.8	Radio + baseband
Network (4G/5G)	0.5 – 2.0	Throughput & signal quality
Navigation	0.2 – 0.6	GNSS sampling
Background	0.2 – 1.0	Number & type of services

Table 2: Estimated power parameters for modern flagship smartphones.

5 Numerical Simulation and Dynamic Performance Analysis

Numerical Analysis and Model Outputs **(i) Continuous-time SOC trajectories and nonlinear discharge behavior** (Section 5.1). We numerically integrate the coupled ODE system to obtain **time-continuous SOC trajectories**, denoted as $\mathrm{SOC}(t)$. This approach captures **intrinsically nonlinear discharge effects**, including rate-dependent capacity loss, recovery phenomena, and curvature in the voltage–SOC relationship. By explicitly resolving the coupling between instantaneous power demand and battery state evolution, the model goes beyond linear or piecewise-linear abstractions and reveals how transient loads translate into macroscopic SOC depletion.

(ii) Component-level power decomposition (Section 5.2). The total power demand is decomposed into **component-wise contributions** from major subsystems such as the CPU, display, radio interfaces, and background services. Evaluating the ODE framework with disaggregated power terms enables a **quantitative attribution of energy consumption**, identifying dominant energy sinks under specific usage patterns. This decomposition directly supports targeted power-management and system-level optimization.

(iii) Time-to-empty (TTE) prediction across scenarios (Section 5.3). Based on the integrated SOC dynamics, we compute the **time-to-empty (TTE)**, defined as the time required for the battery to reach a predefined depletion threshold. By varying both **initial SOC levels** and **usage scenarios**, the model yields a mapping from operating conditions to expected battery lifetime. This enables systematic evaluation of endurance and sensitivity to high-power demand episodes.

(iv) Stochastic Monte Carlo extension (Section 5.4). The deterministic framework is extended to a stochastic setting via **Monte Carlo simulation**, in which usage patterns and selected model parameters are treated as random variables. Repeated numerical integration of the ODE system produces a **distribution of TTE outcomes**, rather than a single point estimate. This probabilistic view captures variability, uncertainty, and tail behavior, providing a more realistic assessment of battery endurance under heterogeneous real-world usage.

Collectively, these four components establish a closed numerical pipeline that links **continuous-time battery dynamics** to **component-level power mechanisms**, translates them into **actionable endurance metrics**, and embeds the analysis within a **stochastic framework** reflecting real-world variability.

5.1 Continuous-Time SOC Trajectories and Discharge Dynamics

We simulated **SOC trajectories** for **five representative usage scenarios** spanning the spectrum of smartphone operation: **Deep Sleep (S1)**, **WiFi Video (S2)**, **4G Social (S3)**, **5G Navigation (S4)**, and **Heavy Gaming (S5)**. For each scenario, simulations were conducted at **multiple initial charge levels** (**100%, 80%, 60%, 40%, and 20%**). Figure 2 plots **remaining SOC versus runtime**. **Solid curves** correspond to the **physics-based closed-loop simulation**, while **dashed lines** indicate **idealized linear discharge** for reference. .

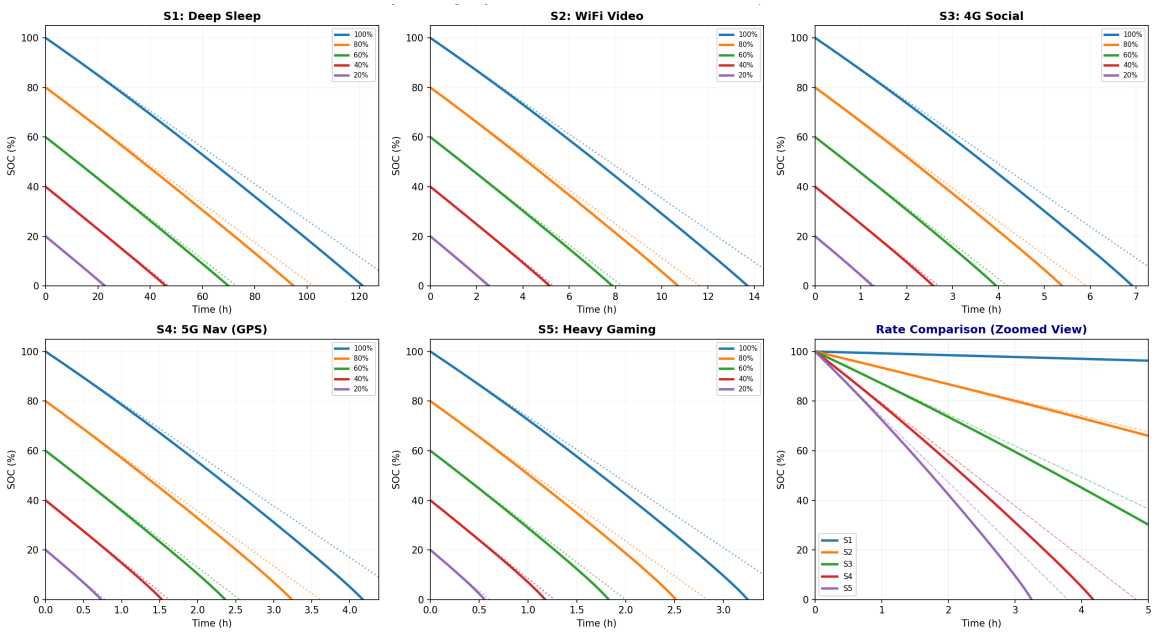

Figure 2: Continuous-time SOC discharge trajectories under five usage scenarios and multiple initial charge levels. Solid lines show physics-based simulations; dashed lines indicate linear reference discharge.

5.1.1 Nonlinear discharge dynamics across scenarios

Across all five scenarios, SOC decreases in a convex, non-linear manner rather than following an idealized linear trajectory. The deviation from linear discharge becomes most pronounced at low SOC (below approximately 20%), indicating accelerated depletion near end-of-life. This behavior reflects the intrinsic closed-loop electrochemical mechanism of lithium-ion batteries: as SOC declines, terminal voltage drops and internal resistance increases; under approximately constant power demand, higher current is required, which amplifies resistive losses and further accelerates voltage collapse.

While the curve shapes are similar, their temporal scales differ dramatically. Total runtime ranges from over 120 hours in Deep Sleep (S1) to approximately 3.3 hours in Heavy Gaming (S5), implying more than an order-of-magnitude difference in effective load. Intermediate scenarios isolate the energetic cost of connectivity and sensing: moving from WiFi Video to cellular social usage and then to 5G Navigation substantially reduces runtime, highlighting the strong impact of cellular radios and GPS.

5.2 Component-Level Power Decomposition and Analysis

To identify the drivers behind the discharge rates observed in Section 5.1, we decompose total power consumption into hardware subsystems. Figure 3 shows the relative contributions of Base, Screen, CPU/GPU, Network, GPS, Background and other components, with titles indicating the total instantaneous power P_{total}.

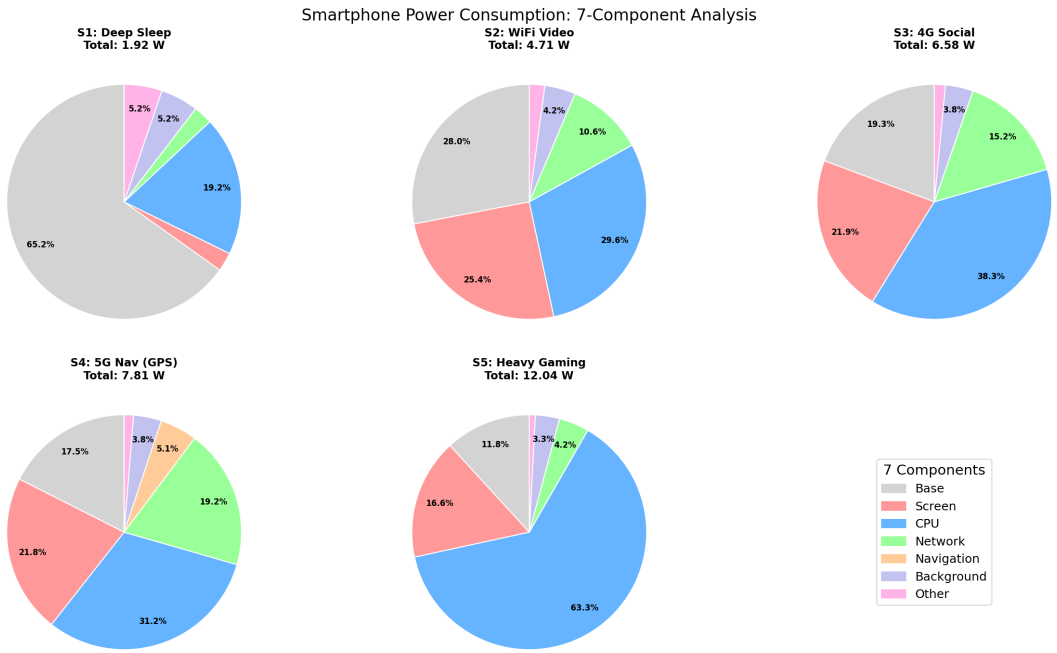

Figure 3: Component-level power decomposition across five usage scenarios. Percentages indicate relative energy contribution; titles show total instantaneous power.

5.2.1 Dominant contributors and load shifts

Absolute power demand varies enormously across scenarios (e.g., ~1.92 W in Deep Sleep vs. ~12.04 W in Heavy Gaming), directly explaining the large runtime disparities. More importantly, the dominant energy sink shifts with context: in Deep Sleep (S1), base and CPU activity dominate, indicating that standby cost is largely the price of connectivity. During active use (S2–S3), the display becomes a leading consumer, while the transition from WiFi to cellular substantially increases total power despite similar percentage shares, revealing the high energy cost of mobile data transmission.

5G Navigation (S4) combines multiple high-demand modules simultaneously (screen, radio, and GPS), creating a compounded load. Heavy Gaming (S5) is characterized by CPU/GPU dominance, which overwhelms other components even though the display and network remain active. Finally, the System component behaves as a near-fixed overhead: it is noticeable at low total load but becomes negligible in percentage terms under heavy workloads.

5.3 Time-to-Empty (TTE) Prediction under Varied Scenarios

We aggregate simulation outputs to predict time-to-empty (TTE) across initial SOC levels. Figure 4 presents TTE on a logarithmic scale to accommodate the large endurance gap across scenarios.

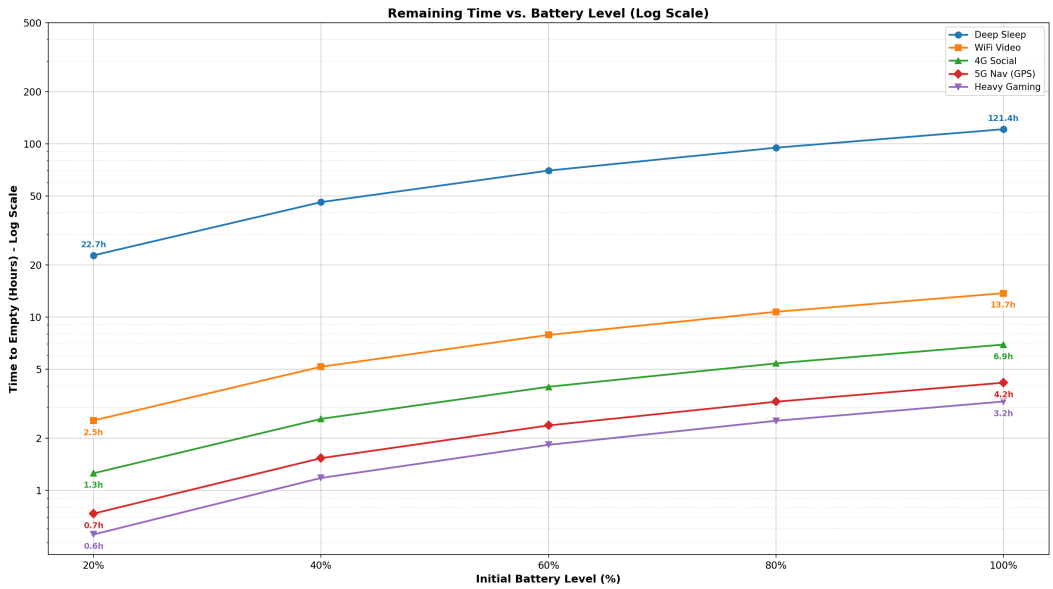

Figure 4: Predicted time-to-empty (TTE) as a function of initial SOC under five usage scenarios, shown on a logarithmic scale.

5.3.1 Scaling behavior and critical thresholds

On the log scale, TTE exhibits a strict hierarchical ordering: at full charge, Deep Sleep provides over 120 hours while Heavy Gaming lasts only about 3 hours, indicating orders-of-magnitude differences in endurance. Curves are approximately parallel, suggesting near-proportional scaling between TTE and initial SOC, with no intersections across scenarios.

At the commonly perceived "low battery" threshold of 20% SOC, remaining runtime varies sharply by load intensity: low-load operation retains substantial usable time, whereas high-load scenarios collapse to less than an hour, quantitatively explaining range anxiety. The marginal utility of additional capacity also diminishes as load increases: raising SOC from 20% to 100% yields nearly 100 extra hours in Deep Sleep but only a few hours under Heavy Gaming, motivating component-level power optimization rather than capacity expansion alone.

5.4 Stochastic Extension via Monte Carlo Simulation

Real-world smartphone usage is inherently stochastic: user interactions fluctuate, network conditions vary, background processes awaken intermittently, and temperature may change. Consequently, even under the same scenario label, TTE should be treated as a random variable. We therefore extend the deterministic framework using Monte Carlo simulation by randomizing time-varying inputs (CPU utilization, brightness, network type and quality, GPS, background activity, and temperature) while preserving the same closed-loop battery dynamics and cutoff condition ($V \leq V_{\text{cut}}$).

5.4.1 Framework and distributional endurance

Figure 5 presents the empirical survival curve $P(\text{TTE} \geq t)$ aggregated over Monte Carlo runs, emphasizing that endurance is best described as a distribution. Scenario-conditioned survival curves (Figure 6) preserve the deterministic ordering while exhibiting substantial dispersion, which quantifies risk: unfavorable combinations of network quality, computational spikes, and background activity can significantly shorten runtime.

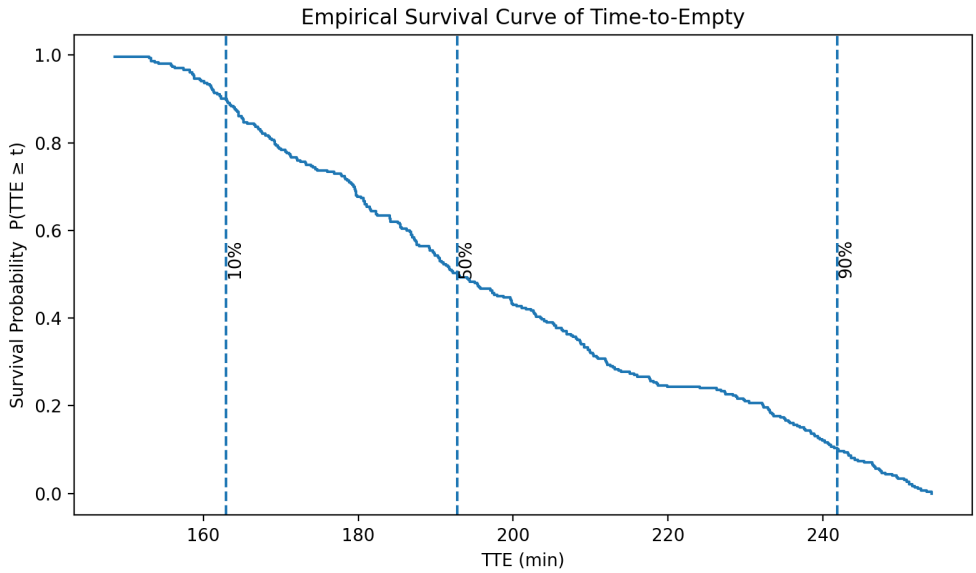

Figure 5: Empirical survival curve of time-to-empty (TTE) obtained from Monte Carlo simulation. Vertical lines indicate representative quantiles, illustrating the distributional nature of battery endurance.

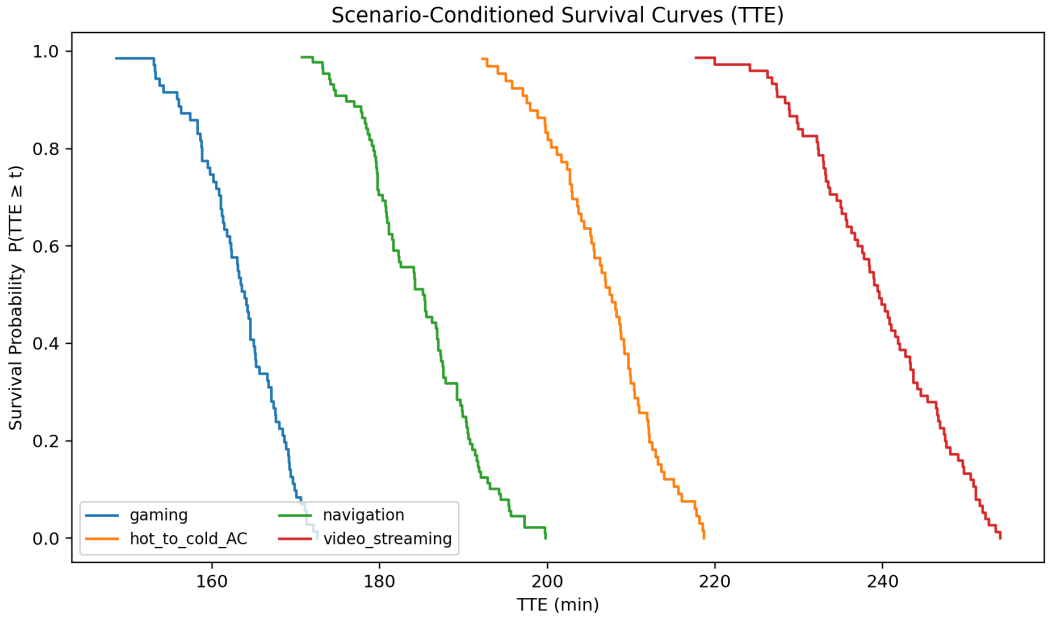

Figure 6: Scenario-conditioned survival curves of TTE. While the hierarchical ordering across scenarios is preserved, each curve exhibits substantial dispersion, reflecting real-world variability.

5.4.2 Energy shares, extremes, and trajectory-level evidence

Monte Carlo aggregation also yields robust energy statistics. Figure 7 reports average energy share by component, confirming the same structural drivers observed in Section 5.2 while smoothing incidental fluctuations. To interpret tail outcomes, we compare the worst 5% and best 5% runs by TTE (Figure 8); worst cases typically combine higher CPU load and brightness with poorer signal quality and larger resistive losses. Finally, representative voltage trajectories and the voltage–SOC phase portrait (Figure 9 and the SOC–voltage phase diagram in Figure 10) show that all runs follow a common voltage-sag manifold and terminate at V_{cut}. Thus, uncertainty in TTE arises primarily from how quickly trajectories traverse the same underlying physical manifold, rather than from changes in battery dynamics.

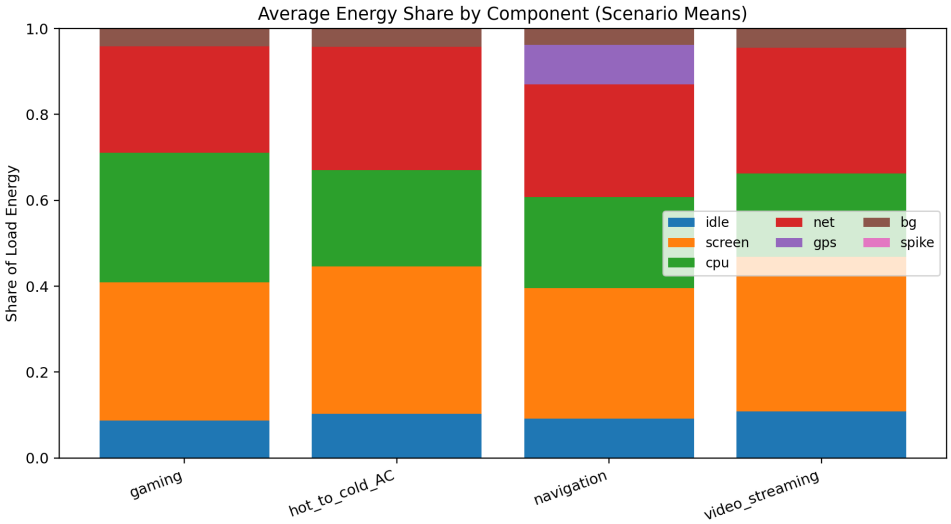

Figure 7: Average energy share by hardware component under stochastic simulation, aggregated by scenario.

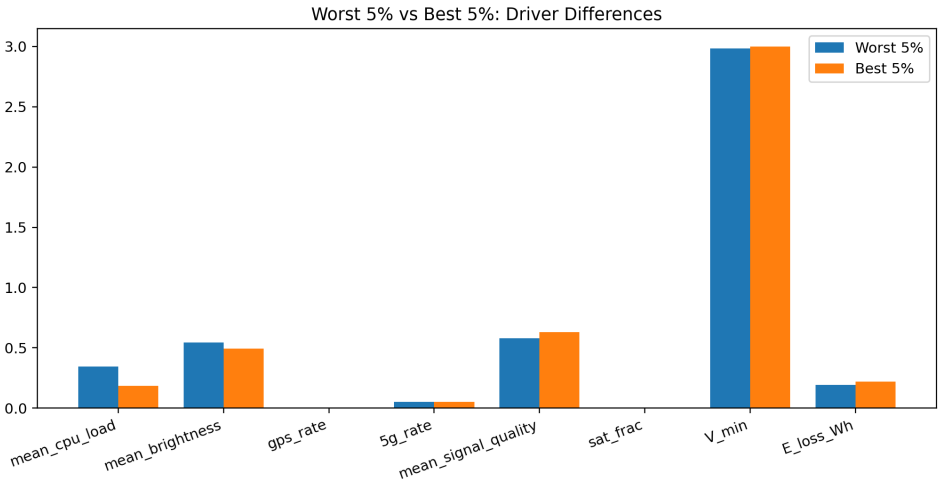

Figure 8: Comparison of key drivers between the worst 5% and best 5% Monte Carlo runs ranked by TTE.

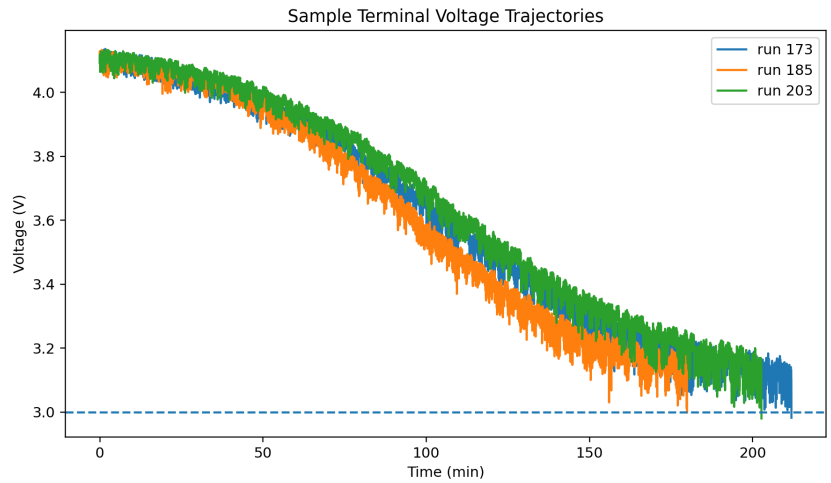

Figure 9: Sample terminal voltage trajectories under stochastic simulation, illustrating voltage sag and cutoff behavior.

6 Sensitivity Analysis and Model Assumptions

6.1 Voltage-Limited Shutdown Mechanism

Smartphone batteries do not discharge until chemical capacity is fully exhausted. Instead, shutdown occurs when the terminal voltage falls below a critical cutoff threshold, even if a non-negligible state of charge (SOC) remains.

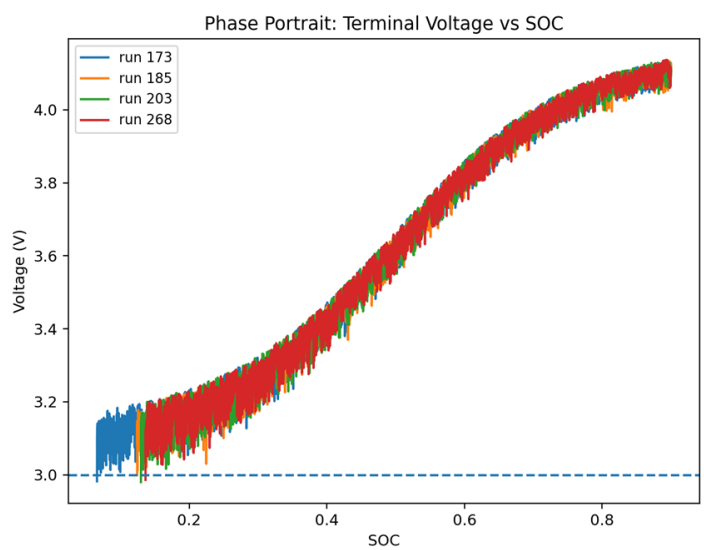

Figure 10: SOC–voltage phase diagram illustrating voltage-limited shutdown behavior under varying load conditions.

Figure 10 illustrates this voltage-limited mechanism via the SOC–voltage phase diagram. As SOC decreases, open-circuit voltage declines gradually, while internal resistance induces an additional voltage drop under load. At low SOC, this interaction produces accelerated voltage collapse, especially under high-power operating

conditions. Consequently, parameters that influence current draw or internal resistance can significantly shift the timing of voltage cutoff, leading to large variations in time-to-empty (TTE).

6.2 Sensitivity Analysis Framework

To assess model robustness and identify dominant drivers of battery lifetime, we perform a one-at-a-time (OAT) tornado sensitivity analysis combined with Monte Carlo simulation. A baseline Monte Carlo experiment using nominal parameters yields a mean TTE of approximately 198 minutes. Each selected parameter is then independently perturbed by ±10%, while all others are held fixed, and the resulting change in mean TTE is recorded.

This approach enables direct comparison across heterogeneous mechanisms, including computational load, internal resistance, network power, thermal effects, and battery aging.

6.3 Tornado Sensitivity Results

Figure 11 presents the resulting tornado diagram, where longer bars indicate greater sensitivity of mean TTE to parameter perturbations.

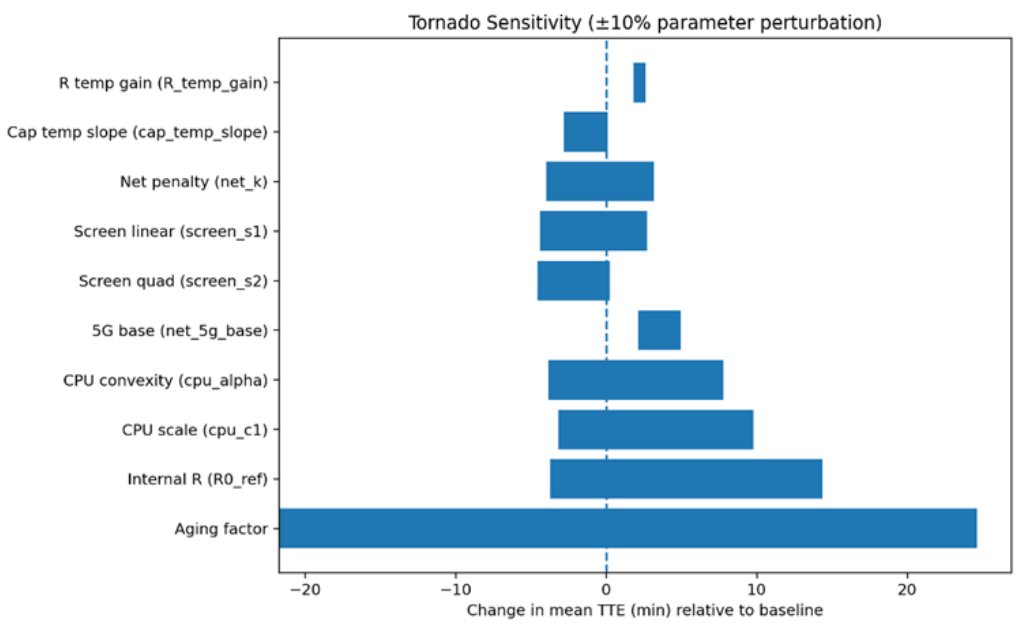

Figure 11: Tornado sensitivity diagram showing changes in mean TTE under ±10% one-at-a-time parameter perturbations.

6.3.1 Dominant and Moderate Parameters

The **battery aging factor** exhibits the **largest overall sensitivity** because it directly scales the **effective usable capacity**, resulting in an approximately **linear impact on runtime**. As this parameter represents **long-term degradation** rather than short-term usage variability, it serves as a **reference scale** for interpreting the relative magnitude of other sensitivities.

The **baseline internal resistance** is the **most influential short-term parameter**. Even small resistance variations significantly shift the **voltage cutoff timing** under high-load conditions, consistent with the **voltage-limited shutdown mechanism** illustrated in Figure 10. **CPU-related parameters** also exhibit strong sensitivity, identifying **computational workload** as a primary driver of short-term battery depletion.

Screen brightness and **network-related parameters** form a **moderate sensitivity tier**, typically inducing runtime changes on the order of **several minutes** under $\pm 10\%$ perturbations. **Temperature-dependent parameters** show comparatively **weaker effects** within the modeled ambient range, although their influence is expected to **increase under extreme thermal conditions**.

6.4 Interpretation, Assumptions, and Robustness

The **asymmetry** observed in many tornado bars reflects **nonlinear interactions** between **current draw**, **internal resistance**, and **voltage cutoff**, rather than modeling artifacts. Importantly, the relative ranking of parameter influence remains **stable across Monte Carlo realizations**, indicating **robust qualitative conclusions**.

Key modeling assumptions include **independent parameter perturbations**, **stationary usage patterns** within scenario blocks, and **typical ambient temperature conditions**. While appropriate for the present scope, these assumptions naturally motivate extensions such as **coupled aging–resistance evolution** and **extreme thermal stress testing**.

6.4.1 Justification of Model Assumptions for Logical Closure

The assumptions introduced in Section 3 are not absolute simplifications but are essential to achieving logical closure in a **continuous-time, physics-based** battery model. They establish a necessary balance between electrochemical fidelity, analytical tractability, and empirical applicability, thereby linking theoretical principles with system-level implementation.

The temperature-dependence assumption is motivated by **the Arrhenius-type kinetics** that govern electrochemical reactions and ion transport. Restricting thermal effects to internal impedance and reaction rates captures the dominant first-order influence of **temperature** on discharge dynamics and aging, while avoiding the prohibitive complexity of a fully coupled thermal–electrochemical model. This ensures that thermal coupling is incorporated where it matters most for state evolution.

The assumption of exponential capacity degradation is grounded in extensive empirical evidence on lithium-ion aging behavior. Representing **long-term capacity** fade through a smooth exponential decay enables the inclusion of aging dynamics within a continuous-time ODE framework, while intentionally excluding rare, stochastic failure modes that lie outside the model's intended operating regime.

Assuming stable statistical properties for power consumption is a prerequisite for predictive capability. Component-level power profiles derived from measurements and benchmarks exhibit **consistent scenario-dependent trends**, allowing user behavior to be mapped to representative power functions. This transforms the model from a purely mechanistic description into a tool capable of simulating real-world variability through parameter distributions and Monte Carlo analysis.

The voltage simplification assumption reflects an **engineering trade-off** rather than a physical omission. While the full terminal voltage model is retained in the coupled state equations, adopting an average voltage for higher-level energy and time-to-empty estimations reduces nonlinear complexity without materially affecting predictive accuracy under normal operating conditions.

Taken together, these assumptions define the **boundary of applicability** of the model and enable the formulation of **a closed, four-state coupled ODE system** that is both physically interpretable and computationally manageable. They are therefore integral to the internal consistency of the study, ensuring that the resulting predictions remain robust within the domain of typical smartphone usage.

6.5 Summary of Sensitivity Findings

Overall, short-term battery life is governed primarily by internal resistance and computational workload, while battery aging dominates long-term capacity loss. These findings confirm the physical consistency of the proposed model and provide a solid foundation for subsequent user-level and system-level recommendations.

7 Insights and Recommendations

Drawing upon the physics-based continuous-time model and the sensitivity analysis presented in the preceding sections, we translate our theoretical findings into actionable strategies. These insights are divided into **practical behavioral adjustments for smartphone users** and **algorithmic optimizations for operating system (OS) power management**, aiming to maximize both the daily "time-to-empty" (TTE) and the long-term lifespan of the battery.

7.1 Practical Recommendations for Users

Our component-level power decomposition reveals that **user behavior significantly influences the non-linear dynamics of battery depletion**, particularly in critical states. The most immediate behavioral adjustment concerns usage patterns at low states of charge. Our discharge trajectory analysis demonstrates that **when the SOC drops below 20%, internal resistance and polarization voltage increase non-linearly**, making the system highly susceptible to "voltage collapse." Consequently, users are strongly advised to **avoid high-power activities at low SOC**, such as 3D gaming or continuous GPS navigation. High current demands in this region amplify resistive losses, causing the terminal voltage V_{term} to prematurely hit the cutoff threshold V_{cut} even if theoretical capacity remains.

Furthermore, connectivity and display settings offer substantial leverage for energy conservation. Simulation results indicate that the **transition from Wi-Fi to cellular data (4G/5G) induces a significant spike in baseband power consumption** (P_{net}), suggesting that users should prioritize Wi-Fi connections for data-intensive tasks like streaming. Regarding display management, the **superlinear scaling of power with brightness** implies that dimming the screen from a maximum setting yields disproportionately larger savings than adjustments at lower levels. Thus, utilizing auto-brightness to avoid static high-luminance states is a highly effective strategy. Finally, governed by the Arrhenius law in our aging model, users should **minimize device exposure to high ambient temperatures**, especially during charging, to retard the exponential acceleration of capacity fade (f_{age}).

7.2 Strategies for Operating System Optimization

Beyond user behavior, the proposed coupled ODE framework offers **white-box insights that can be integrated directly into OS power management policies**. We propose shifting from purely SOC-based power saving modes to a **predictive, voltage-aware throttling mechanism**. A "Survival Mode" should be triggered not merely when the battery percentage is low, but **when the model predicts a high risk of voltage collapse**. By dynamically capping the maximum CPU frequency $f(t)$ and screen brightness based on the real-time proximity of V_{term} to V_{cut}, the OS can **smooth the discharge curve and prevent sudden shutdowns**, effectively extending the usable tail of battery life.

Additionally, the operating system can optimize energy efficiency through **context-aware background scheduling**. Since aggregated background tasks contribute a steady baseline power drain, the system should batch synchronization events and restrict them to periods of efficient connectivity, such as when Wi-Fi is available. Long-term stability can further be enhanced by implementing **aging-adaptive thermal management**. As the battery ages and its internal resistance increases, the heat generation rate for a given current rises. Therefore, the

OS should continuously estimate the aging factor f_{age} and **recalibrate thermal throttling limits accordingly**, ensuring safety and longevity throughout the device's lifecycle.

8 Conclusion

In this study, we developed a **physics-based continuous-time framework** to model smartphone battery dynamics, effectively bridging the gap between electrochemical principles and real-world usage patterns. By formulating a coupled system of **four ordinary differential equations (ODEs)**, we captured the intricate interactions between State of Charge (SOC), polarization voltage, internal temperature, and capacity fade. Unlike conventional monolithic approaches, our **component-wise power decomposition** allowed for a granular analysis of energy consumption across display, CPU, and network activities. This structure enabled our model to faithfully reproduce complex non-linear phenomena—most notably the **accelerated voltage collapse observed at low SOCs**—thereby offering a more accurate and interpretable alternative to pure data-driven methods.

Furthermore, our sensitivity analysis revealed that **battery aging is the dominant factor in long-term performance degradation**, while CPU workload and internal resistance primarily govern daily discharge variability. The model's robustness was confirmed across diverse usage scenarios, demonstrating that "time-to-empty" is **governed by the coupling between instantaneous current demand and the battery's voltage state**. Ultimately, these insights provide a theoretical foundation for **voltage-aware throttling and context-sensitive background scheduling**, demonstrating that integrating physical constraints with software management is essential for next-generation power efficiency.

References

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Report on Use of AI Tools

This section documents the transparent and responsible use of AI-assisted tools in the preparation of this submission, in accordance with COMAP's *Use of AI Tools in COMAP Contests* policy.

1. OpenAI ChatGPT (Model: GPT-5.2)

Access dates: January 29 – February 2, 2026

OpenAI ChatGPT (model version GPT-5.2) was used as a **supportive productivity tool** during the preparation of this report. Its role was limited to assisting with implementation, presentation, and language refinement, and it did *not* replace independent mathematical reasoning, modeling decisions, or scientific judgment by the team.

Importantly, **all modeling ideas, assumptions, system structure, mathematical formulations, and overall algorithmic design were conceived and determined by the team**. AI assistance was used only after the modeling framework had been established.

The purposes and scope of use are detailed below by date.

January 29, 2026

ChatGPT was used to assist with:

- Discussing high-level modeling structure after the core framework was designed by the team
- Improving clarity and academic tone of preliminary English drafts

All modeling concepts and assumptions originated from team discussions.

January 30, 2026

ChatGPT was used to assist with:

- Providing implementation-level assistance for coding, such as syntax clarification and modular code organization
- LaTeX-related questions, including equation formatting and figure placement

The **algorithmic logic, parameter definitions, and computational workflow** were specified by the team prior to implementation.

January 31, 2026

ChatGPT was used to assist with:

- Supporting the translation of established modeling ideas into executable code
- Suggesting improvements to code readability and structure
- Drafting explanatory text describing Monte Carlo simulation procedures

All simulation code was reviewed, tested, and validated by team members.

February 1, 2026

ChatGPT was used to assist with:

- Minor debugging support and error interpretation during code implementation
- Refining explanations of sensitivity analysis results

Numerical experiments, result verification, and interpretation were carried out independently by the team.

February 2, 2026

ChatGPT was used to assist with:

- Final proofreading and language polishing of the complete manuscript
- Formatting adjustments in LaTeX to improve presentation quality

No new modeling logic or computational structure was introduced at this stage.

2. Coding Responsibility and Verification

While AI tools provided assistance during the coding process, **the overall computational framework, simulation design, parameterization, and validation strategy were fully developed by the team**. AI-assisted code suggestions were treated as drafts and were modified as necessary to ensure correctness, efficiency, and consistency with the intended model.

All final code used in simulations was:

- Fully understood by team members
- Manually reviewed and tested
- Executed and validated using independent checks

3. Statement of Responsibility

The team takes full responsibility for the originality, correctness, and integrity of all modeling decisions, mathematical derivations, computational implementations, results, and conclusions presented in this report. AI tools were used solely as assistive technologies and not as substitutes for human creativity, critical thinking, or scientific judgment.